

Supplementary Material

This supplementary material documents implementation details for the GrapeMaster software platform. The main manuscript presents design requirements, architecture, analytical-model integration, user workflows, and a deployment demonstration. This document provides identifier relationships, service-interface and evidence boundaries, workflow reconstruction rules, the software stack, selected data-model attributes, and operational-record screening rules for technical inspection.

Supplementary figures

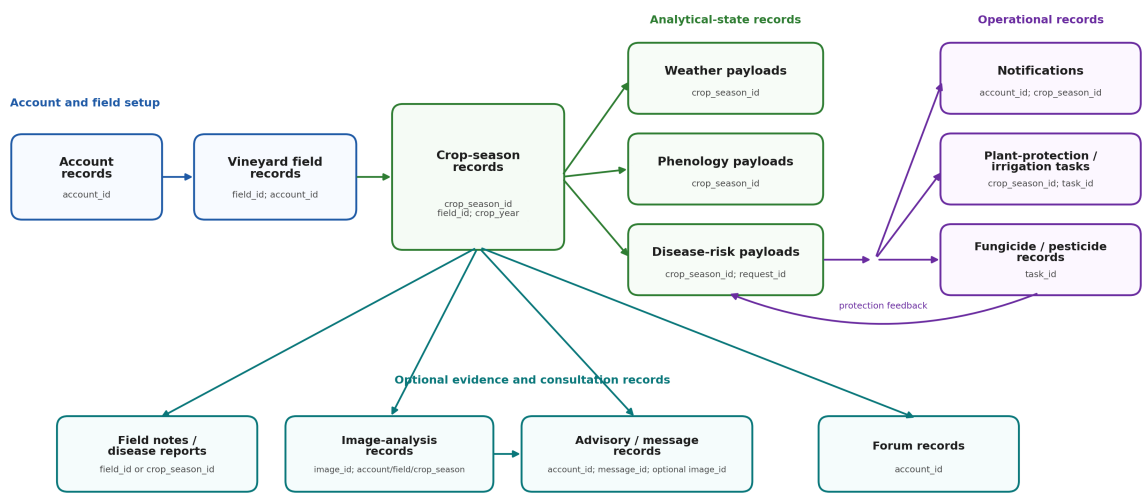


Figure S1. Backend identifier relationships among GrapeMaster record classes. The crop record anchors the principal monitoring and management resources. Lines denote direct or contextual retrieval relationships according to the record type; they should not all be interpreted as direct foreign keys or causal provenance.

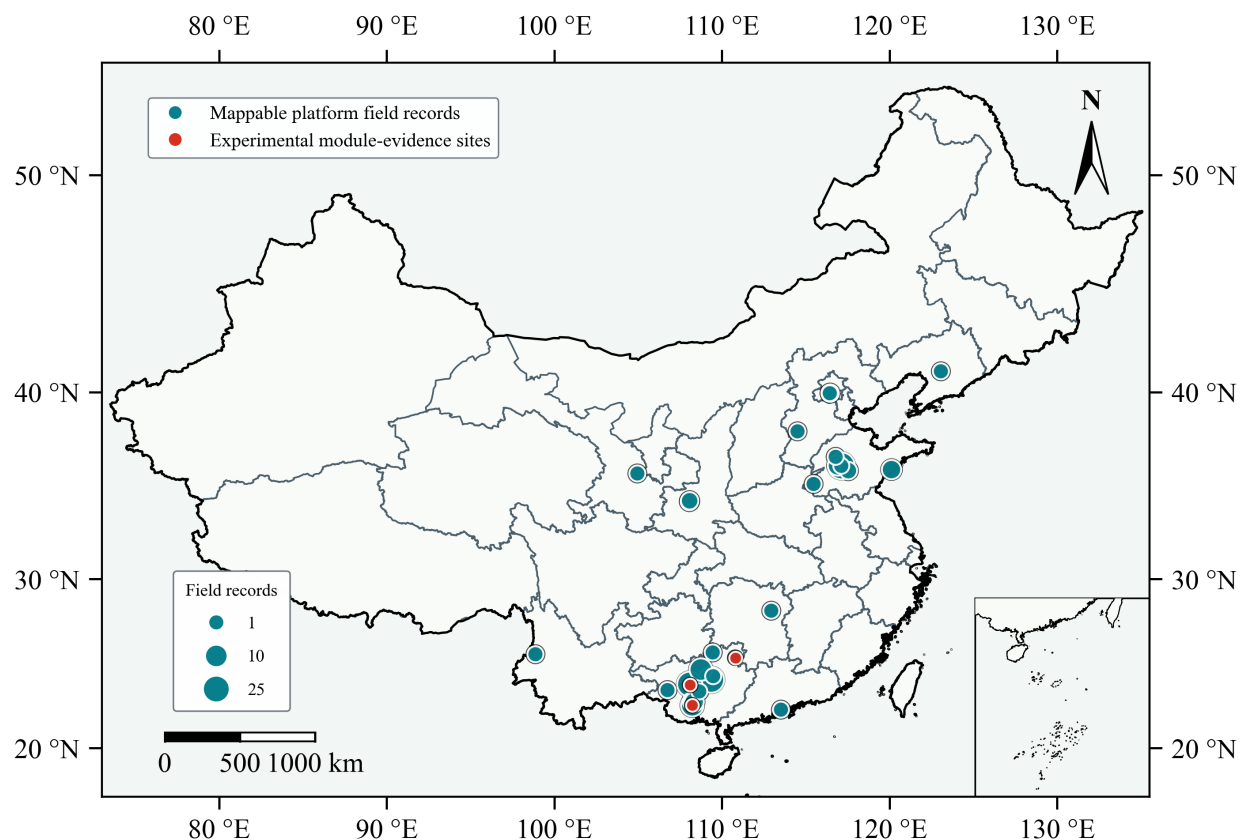


Figure S2. Spatial context of mappable platform field records and independent experimental module-validation sites. Teal bubbles show mappable field records aggregated by region, with bubble size indicating the number of records per region. Red points indicate experimental sites used for module-level validation and are not part of the operational-record denominator.

Supplementary tables

Table S1. Input-output definitions and software roles of integrated analytical components.

Service	Inputs and outputs	Invocation, workflow position, and evidence boundary
Weather service	Inputs include field coordinates and requested time windows. Outputs include historical and forecast weather plus derived spray-suitability variables.	The Django backend calls the Agromodel weather and geocoding service during initialization and scheduled refresh. Returned data are transformed and stored under crop-linked resources.

Service	Inputs and outputs	Invocation, workflow position, and evidence boundary
Phenology processing	Inputs include cultivar and crop metadata plus date-aligned daily temperature. Outputs include broad BBCH-related stage states and a daily stage sequence.	Runs as part of backend crop refresh and supplies both mobile display and host-state input for disease processing. The Guangxi result in the main paper supports the configured component.
Disease-risk service	Inputs include hourly weather, growth stage, cultivar susceptibility, target disease, and recent applied-fungicide history. Outputs include disease states, field risk, protection interpretation, recommendations, and treatment windows.	The stateless FastAPI service is called by Django; request and response are retained by the backend. The current source contains downy and powdery mildew paths. The FSIM-S result supports one platform-compatible Guangxi configuration but is not tied to every deployment record by an immutable version identifier.
Image-recognition service	Input is a user-selected symptom image. Output is hosted-service classification text and confidence information.	Called directly by the Flutter client as a post-symptomatic service. The service source and deployed checkpoint are outside the principal backend repository; seven-class performance is reported as supporting model evidence.
LLM-assisted advisory service	Inputs are user questions and diagnostic context available to the hosted advisory channel. Outputs are streamed management-oriented text terminated by an end marker.	Called over WebSocket by the Flutter client. The client contract is documented, but the hosted checkpoint, fine-tuning data, prompt, and version were not included in the available source. Advisory quality was not independently evaluated.

Table S2. Database relationship rules supporting crop-season-centered traceability.

Relationship rule	Required linkage	Verification purpose
Account-to-field linkage	Retained account identifiers must link to at least one vineyard field record	Confirms that the account has an operational vineyard context rather than registration-only activity
Field-to-crop-season linkage	Each evaluated crop-season record must be linked to a field record through field or crop-season identifiers	Separates persistent field identity from season-specific analytical and management state
Crop-season-to-analytical-payload linkage	Weather, phenology, disease-risk, and model request-response payloads should share the crop-season identifier where applicable	Confirms that analytical inputs and outputs can be retrieved under the same seasonal anchor
Crop-season-to-event linkage	Notifications, warning events, and task records may be associated through retained user, field, or crop context	Supports contextual retrieval; a task does not store a dedicated source-notification foreign key, so event-to-task causality is not established
Task-to-product linkage	Fungicide or pesticide product records must link to plant-protection tasks through task identifiers	Confirms that completed operations can be converted into product-level treatment feedback
Crop-season-to-symptom/advisory record linkage	Image records, advisory records, field notes, and disease reports use the account, field, crop, image, or message identifiers available to each function	Supports functional association while acknowledging that advisory messages are primarily user- or message-keyed rather than uniformly crop-keyed
Explicit absence rule	If an expected optional record type is absent for a replay anchor, the absence is recorded rather than inferred as failure of unrelated workflow steps	Prevents overclaiming and separates missing symptom or advisory records from successful risk-task-feedback linkage

Table S3. Rules used for the deployment-record demonstration and representative workflow reconstruction.

Verification check	Operational criterion	Supported interpretation
Operational-account retention	An account is retained only when it has business records and field-crop-season linkage after deduplication and target-context screening	Defines the evaluated operational account set
Analytical payload completeness	A crop season is counted as analytically complete when weather, phenology, disease-risk, and request-response payloads are present under the shared seasonal context	Shows whether the analytical state can be reconstructed for that season
Disease-risk-to-operation context	Notification and task records are checked against retained user, field, or crop context	Shows whether event and operation objects coexist in the workflow context; it does not prove that one event caused one task
Treatment-feedback linkage	Fungicide or pesticide records are checked through task identifiers and task completion status	Shows whether field operations can be converted into feedback records
Symptom/advisory record-linkage availability	Image-analysis, advisory, disease-report, and note records are checked for available account, field, crop-season, image, or message links	Shows whether symptom-related records and advisory interactions are available for seasonal interpretation
Representative workflow	A selected crop anchor retrieves weather, phenology, disease-risk, notification, task, and fungicide records through shared crop or task keys	Illustrates the implemented risk-to-action-to-feedback objects and their available linkage strengths
Boundary reporting	Missing optional symptom or advisory records under the replay anchor are explicitly reported rather than omitted	Separates verified linkage from unsupported production-outcome claims

Verification check	Operational criterion	Supported interpretation
Excluded outcome claim	Operational-record linkage and replay checks are not treated as demonstrations of pesticide reduction, disease-control efficacy, economic benefit, or user adoption	Maintains the distinction between record-based workflow assessment and agronomic-impact evaluation

Table S4. Record-level reconstruction results for a representative GrapeMaster crop context.

Record group	Retained record set	Identifier or status	Result note
Replay record set	Weather, phenology, disease-risk, notification, task, and fungicide records	One crop UUID	Representative workflow record set
State and risk records	Weather, phenology, and disease-risk payloads	Shared crop-season key	Shared seasonal record set
Operation records	Notifications and completed plant-protection tasks	Same crop context, without a direct source-notification task key	Contextual event and operation records
Treatment-feedback records	Fungicide-product records	Task identifiers	Product-task record set
Field-note branch	No field-note or incidence-note record	Explicit absence	No field/incidence note in this replay

Table S5. Software stack and implementation components used in the GrapeMaster platform.

Layer	Component group	Main technologies or libraries
Mobile client	Application framework	Dart, Flutter
Mobile client	Android shell and build configuration	Kotlin, Gradle

Layer	Component group	Main technologies or libraries
Backend business service	Web framework and API layer	Python, Django 4.2, Django REST Framework
Backend business service	Authentication, scheduling, and asynchronous communication	SimpleJWT, Django APScheduler, Django Channels
Backend business service	Database access and persistence	PostgreSQL, psycopg2, normalized relational tables, JSON model payloads
Disease-model service	Service interface	FastAPI, Uvicorn, Pydantic, Starlette
Weather and time-series processing	Numerical and date-time processing	NumPy, Pandas, Python Dateutil, Pytz
Weather and geospatial processing	Field geometry, geocoding, and weather-service access	Shapely, Fiona, GeoPandas, Geopy, Requests, HTTPX, Agromodel weather/geocoding API, request caching, retry utilities
Phenology and risk analytics	Crop-stage and risk-state calculation	Python analytical workflows, date-aligned weather transformation, cultivar and crop-season metadata
Image processing and recognition	Image preprocessing and deep-learning inference	PyTorch, TorchVision, OpenCV, Pillow, Ultralytics-related components, EfficientViT-related modules, segmentation components
Advisory interface	Text advisory channel	Hosted WebSocket advisory service used by Flutter; concept follows the published CDIP-ChatGLM3 framework, while the deployed model artifact is externally maintained

Table S6. Selected data-model attributes supporting the crop-season-centered workflow.

Record group	Main attributes or stored content	Platform role
Field record	User linkage, field name, area, GeoJSON boundary, centroid coordinates, administrative region, sharing state, deletion state, notice state, and generated field-boundary preview	Persistent spatial and ownership unit for vineyard management
Field geometry processing	Boundary parsing, boundary-image generation, area estimation, centroid calculation, and optional geocoding request	Converts user-defined field boundaries into location and display objects
Crop-season record	UUID, field linkage, grape cultivar, cultivation method, vine age, field coordinates, field size, cultivar metadata, and timestamps	Field-linked crop-management context for weather, phenology, risk, notifications, tasks, and products; explicit season year, start, end, and lifecycle state are not implemented
Master-data records	Grape variety, cultivar susceptibility, cultivation method, BBCH stage, disease type, fungicide, pesticide, and reminder rules	Controlled reference values for field, crop-season, task, risk, and advisory records
Model payload records	Weather, phenology, disease-risk, and request-response payloads stored with structured records	Preserves inputs and outputs for mobile display, recalculation, and audit

Table S7. Operational-record screening rules used before summarizing deployed GrapeMaster records.

Screening step	Rule	Rationale
User-record integration	User records were integrated using available user identifier, custom-user identifier, and phone-number fields.	Avoids duplicate counting of account-level records

Screening step	Rule	Rationale
Business-record filter	Accounts with no business records across field, crop-season, notification, task, fungicide or pesticide, advisory or message, forum, disease-report, and note tables were excluded.	Removes registration-only accounts from operational summaries
Field-crop-season linkage filter	Accounts without both field records and crop-season records were excluded.	Retains records that contain the principal setup objects used in the deployment demonstration
Target-context filter	A single account associated only with Linwei District was excluded.	Retains the intended vineyard deployment context for the descriptive record summary
Retained account set	Forty-seven accounts remained from 95 deduplicated accounts.	Defines the candidate account set used for operational-record coverage

Table S8. Verified implementation evidence and remaining archival requirements.

Component	Verified in the documented source and records	Required for a submission-grade archive
Flutter client	Field, crop, weather, risk, task, product, image, report, and streamed advisory flows; fixed service endpoints and parsed response contracts	Versioned release, configurable endpoints, build instructions, sample configuration, and interface-contract tests
Django backend	REST resources, field/crop identifiers, scheduled weather refresh, weather/phenology/disease payload storage, notifications, tasks, product linkage, and treatment-history translation	Deployment configuration, database migration snapshot, environment template, scheduler deployment guidance, automated tests, and dependency lock

Component	Verified in the documented source and records	Required for a submission-grade archive
Disease service	FastAPI schema, stateless request execution, downy and powdery pathways, field-risk and recommendation outputs, and fungicide-history input	Immutable model configuration, tagged source release, regression fixtures, deployment-to-validation version mapping, and parameter provenance
Image service	Client HTTP request, result parsing, image-related application records, and supporting seven-class performance result	Hosted source, checkpoint, class mapping, preprocessing, endpoint schema, version identifier, and deployment instructions
LLM advisory	Client WebSocket request and streaming response behavior plus advisory/message records	Model checkpoint, fine-tuning or retrieval data description, prompt and system policy, endpoint schema, version identifier, safety evaluation, and deployment instructions
Operational records	Aggregated counts, crop-context linkage, task-to-product linkage, and one representative reconstruction	De-identification script, schema dictionary, reproducible extraction query, and archived derived demonstration dataset